

In-Class Exercise: Python f-Strings and Unit Conversions

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1 Formatted Strings

1. Python **f-strings** allow numerical values to be inserted into strings and control how those values are displayed.

- Try the following statements in the interpreter:

```
x = 3.14159265
print(f"{x:f}")
print(f"{x:.4f}")
print(f"{x:8.4f}")
print(f"{x:08.4f}")
```

The format **f** uses fixed-point notation, **.4f** displays four digits after the decimal, **8.4f** uses a field at least eight characters wide, and **08.4f** fills unused space with zeros.

- Integer values can be formatted in a similar way:

```
n = 12345678
print(f"{n:d}")
print(f"{n:,d}")
print(f"{n:10d}")
print(f"{n:010d}")
```

Predict the output of each statement before running it.

2 In-Class Exercise

1. Write a Python program called `unit_conversions.py` that calculates and prints a table of British-to-metric unit conversions. Use 1 ft. = 0.3048 m, 1 mile = 1.609344 km, 1 lb. = 0.45359237 kg, 1 stone = 14 lb., 1 kg = 1000 gm, and 1 mph = 1.609344 km/hr. Perform the following conversions:

Input	Convert to
125 ft	meters
3.75 miles	kilometers
175 lb	kilograms
12 stone	kilograms
2.5 lb	grams
0.5 stone	grams
65 mph	kilometers/hour

- Store each starting value in a variable and calculate each converted value using Python arithmetic.
- Distance and speed results must contain **3 significant figures**. For these particular values, use the appropriate fixed-point precision:

```
38.1
6.04
105
```

Thus, different rows may require **.1f**, **.2f**, or **.0f**.

- Weight results must contain **4 significant figures**. For these particular values, the results should have forms such as

```
79.38
76.20
1134
3175
```

Again, select the appropriate number of decimal places.

- Use field widths so that the numerical results line up in columns. For example,

```
print(f"Feet to meters      "
      f"{feet:8.1f}"
      f"{meters:10.1f}")
```

- Your final output should have the general form

Unit Conversion Table

Conversion	Input	Result
Feet to meters	125.0	38.1
Miles to km	3.75	6.04
Lb to kg	175.0	79.38
Stone to kg	12.0	76.20
Lb to grams	2.50	1134
Stone to grams	0.50	3175
Mph to km/hr	65.0	105

Do not type the converted results directly into the `print()` statements. Python must calculate each result before the f-string formats it.